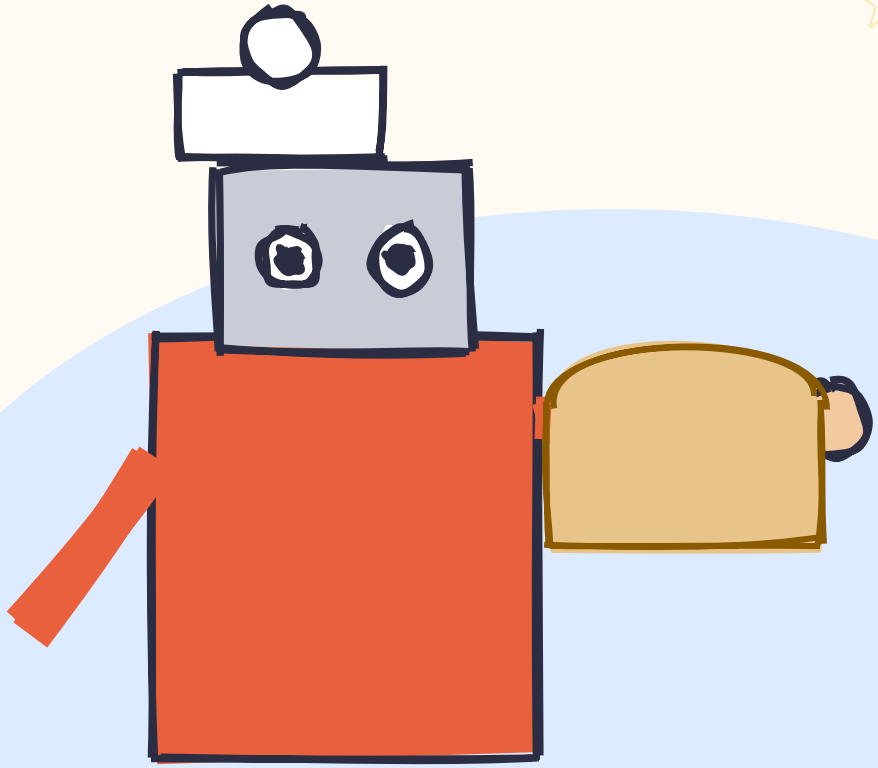
Little Explorers · Class 17

Robot Chef Algorithms

Teach a robot to make a sandwich!

Steps in order **Exact steps** **Fix mistakes**



Did you know?
Robots in some kitchens can flip burgers and make pizza. They follow exact steps called algorithms.

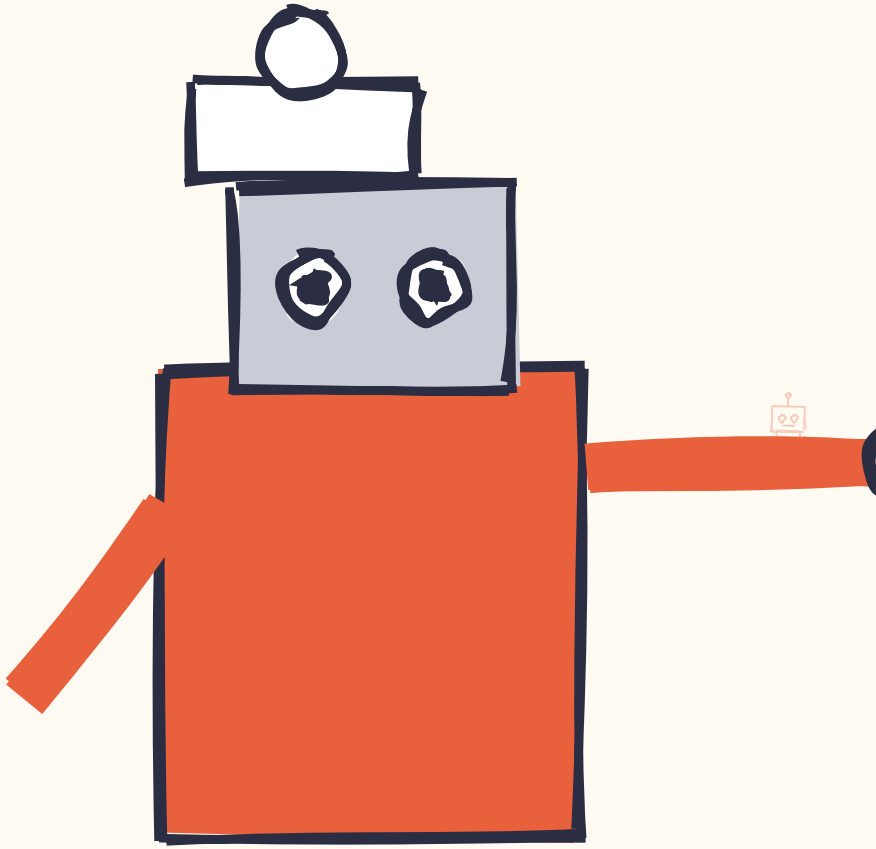
LEARN

Let's wonder...

Can a robot make a jam sandwich?

A hungry robot

Our robot chef is very good at following instructions. But it knows NOTHING about food! If we say 'make a sandwich', it just stands there. What must we tell it?



Make a sandwich!

Be clever!

Every step, in order!

Today: you write a **step-by-step recipe** for a robot chef and test it on a friend.

LEARN

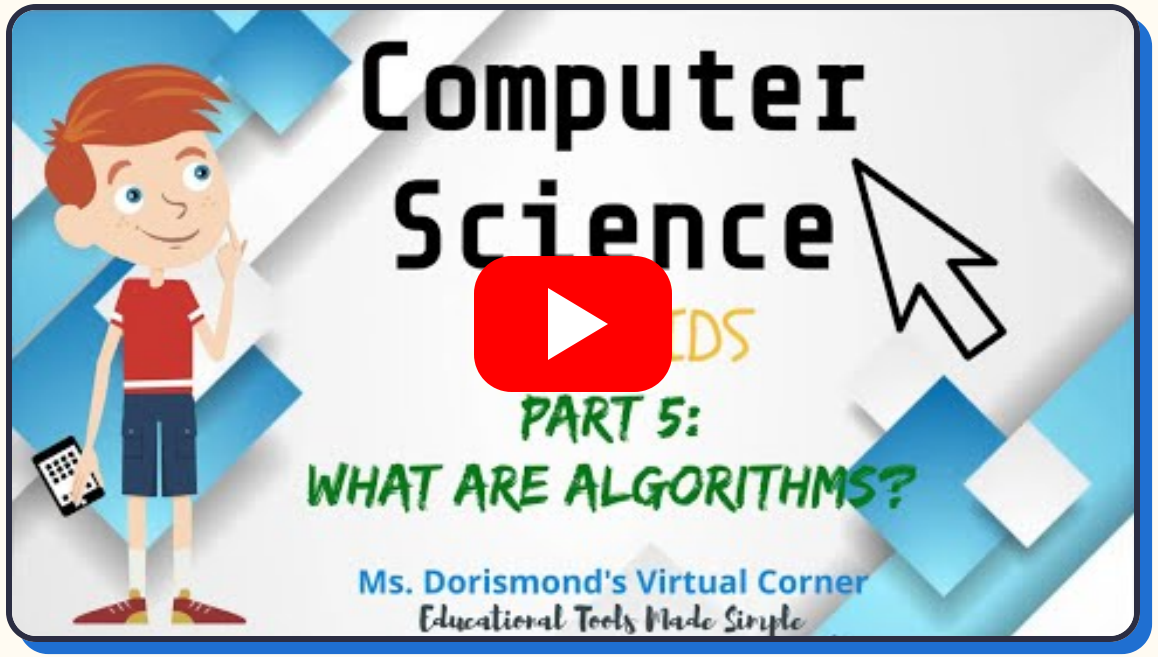
Learning objectives

Today I can...

1	**Steps in order:** put **steps in the right order** to make an algorithm	**Proof:** my picture cards are in the right order
2	**Exact steps:** give **exact** instructions a robot can follow	**Proof:** my robot chef made the sandwich
3	**Fix it:** find and **fix a mistake** in the steps	**Proof:** I fixed one step that was wrong

Objective 1 · Steps in order · video

Video time!



What are Algorithms?

Ms. Dorismond's Virtual Corner · about 3 minutes · youtu.be/578hB0E6y4o

Watch for...

1. What is an **algorithm**?
2. Why must the steps be in the right **order**?
3. Name one algorithm you do every day.

Scan to watch

After the video: Tell your partner the steps to make a jam sandwich. Did you miss any?

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Objective 1 · Steps in order

An algorithm is steps in order

The idea

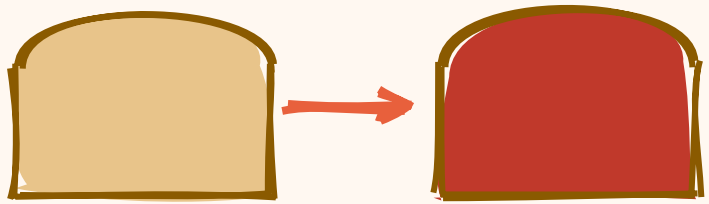
An **algorithm** is a list of steps that you do **one after another** to finish a job.

Getting dressed: socks, then shoes. Not shoes, then socks!

Computers too

Computers follow algorithms all day long, very, very fast!

Example: brushing teeth



1. Get your toothbrush. 2. Put toothpaste on. 3. Brush. 4. Rinse. That's an algorithm! If you rinse first, it doesn't work.

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Objective 2 · Exact steps

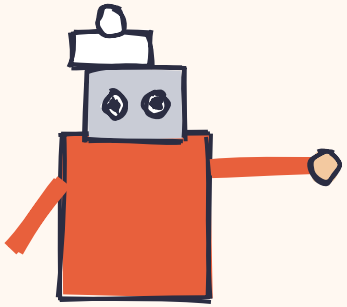
Robots need EXACT steps

The idea

A robot does **exactly** what you say, nothing more.

Say 'put jam on the bread' and it might put the whole jar on the bread! We must say: 'open the jar, take the spoon, scoop jam, spread it on the bread'.

Example: a silly robot



Tell a friend 'walk to the door'. Easy! Tell a robot and it might walk into the wall, because you didn't say 'turn'. Robots need every tiny step.

LEARN

Objective 3 · Fix it

Fixing mistakes: debugging

The idea

When a step is missing or in the wrong place, the robot gets mixed up. That's a **bug**!

To **debug**: read the steps again, find the mistake, change it, try again.

Example: a jigsaw puzzle

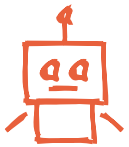


If one puzzle piece is in the wrong place, the picture looks wrong. You don't throw the puzzle away. You find the piece and move it!

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Real world

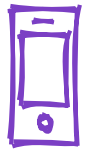
Algorithms everywhere



Kitchen robots
Follow recipes step by step



Washing machines
Fill, wash, rinse, spin



Phones
Every app is made of algorithms



You!
Getting ready for school is an algorithm

? What algorithm do you follow every morning?

Fun fact
A cake recipe is an algorithm that ends in something yummy!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Picture card line-up
2. Our robot chef recipe
3. Be the robot chef!
4. Bug hunt
5. Chef challenges

Class game

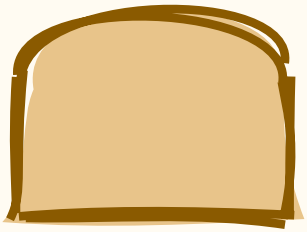
Picture card line-up

How to play

1. Each team gets 5 picture cards: bread, open jar, spread jam, put bread on top, eat!
2. Mix them up.
3. Put them in the **right order** on the table.
4. Check with another team. Are your orders the same?

Then discuss

Was there more than one right order? Which steps MUST come first?



Big idea

The order of steps matters!

MAKE

Objective 2 · Exact steps · write it

Our robot chef recipe

Step	What the robot does
1	
2	
3	
4	
5	
6	

Use action words

pick up · open · scoop · spread · put down · turn

Draw it!
Can't write yet? Draw each step in the box.

MAKE

Class game

Be the robot chef!

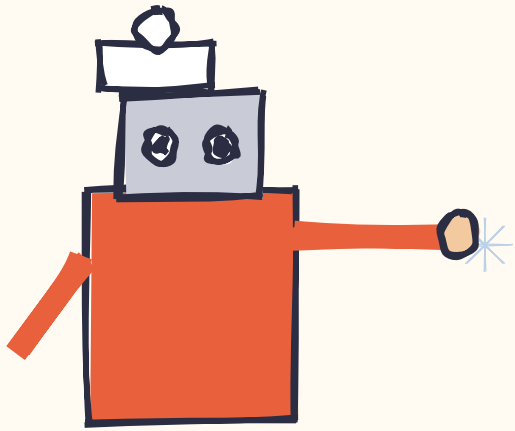
How to play

1. One child is the **robot chef**. Another reads the team's recipe out loud.
2. The robot does EXACTLY what is said, even if it's silly!
3. Use pretend food (paper bread, a cup for the jam jar).
4. Did the robot make the sandwich?

Then discuss

What went wrong? Which step was missing?

Big idea
If the robot gets it wrong, fix the STEPS, not the robot.



MAKE

Debug it!

Bug hunt

Recipe	What's wrong?	Our fix
Spread jam → open jar		
Put bread → eat → spread		
Pick up spoon (no 'scoop')		

Talk about it

Which bug was the funniest?

Which was the hardest to find?

Read out loud

Reading the steps out loud helps you hear mistakes.

MAKE

Level up!

Chef challenges

Choose your challenge



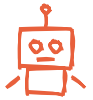
Fruit salad: write the steps to make a fruit salad.



Repeat: 'spread jam' 3 times on 3 pieces of bread. Can you say it shorter?



Swap: follow another team's recipe exactly.



Dance robot: write steps for a robot dance!

Think like an engineer

Why do robots need more steps than people?

Boss level

Write the shortest recipe that still works!

Objective check

Can I really do it? Prove it.

01 Put the 5 sandwich cards in order. ○ ○ ○

02 Read out one EXACT step for the robot chef. ○ ○ ○

03 Tell us one bug you found and how you fixed it. ○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Recipe wall: stick all the team recipes on the wall.

Show what you know

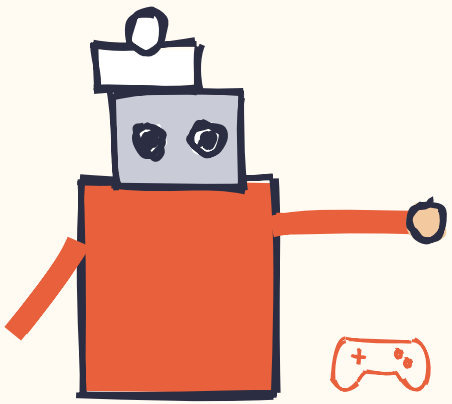
What did we learn?

- [x] An algorithm is steps in order
- [x] Robots need exact steps
- [x] A bug is a mistake in the steps
- [x] Debugging means finding and fixing it

Exit ticket: what is an algorithm?

At home

Tell your family the algorithm for brushing your teeth.



Next class:
Code.org Puzzle Quest!